

Companion LXXII: Kernel PCA of Persistence Diagrams in the Relaxation-Driven Cyclic Cosmology

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April 2026

Abstract

Kernel principal component analysis (kernel PCA) provides a spectral decomposition of persistence diagrams embedded in a reproducing kernel Hilbert space (RKHS). In weak-lensing analyses, this decomposition reveals the dominant modes of topological variation across connected components, loops, and void-like structures. In the standard cosmological model, the kernel spectrum reflects the nonlinear matter distribution and the projection of large-scale structure. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the kernel eigenvalue spectrum in a characteristic way. This Companion develops a continuous description of kernel PCA in RDCC, deriving how the relaxation scale influences spectral topology, principal components, and the observational signatures in weak-lensing surveys. The resulting framework provides a distinctive probe of the RDCC gravitational field through spectral decomposition of linearized topology.

1 Introduction

Persistence diagrams encode the birth and death of topological features in weak-lensing convergence fields. Hilbert-space embeddings provide a linearized representation of these diagrams, enabling kernel methods and inner-product computations. Kernel PCA extends this framework by decomposing the embedded topology into orthogonal spectral modes.

In the standard cosmological model, the kernel eigenvalue spectrum reflects the nonlinear matter distribution and the projection of large-scale structure. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the kernel spectrum in a scale-dependent manner, producing a distinctive signature in kernel PCA analyses.

2 Kernel PCA of Persistence Diagrams

Given persistence diagrams $D_1, \dots, D_N$ embedded into an RKHS via $\Phi(D_i)$, the kernel matrix is

$$K_{ij} = \langle \Phi(D_i), \Phi(D_j) \rangle_{\mathcal{H}}. \quad (1)$$

Kernel PCA solves the eigenvalue problem

$$K \mathbf{v}_k = \lambda_k \mathbf{v}_k, \quad (2)$$

where λ_k are the kernel eigenvalues and $\mathbf{v}_k$ the corresponding eigenvectors.

The k -th principal component is

$$\text{PC}_k(D_i) = \sum_{j=1}^N v_{k,j} K_{ij}. \quad (3)$$

3 RDCC Modification of Kernel Structure

In RDCC, the convergence evolves as

$$\kappa_{\text{RDCC}}(\ell) = \kappa(\ell) \left[1 - \chi_{\kappa} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (4)$$

This modifies the birth and death thresholds of topological features, and hence the kernel:

$$K_{ij}^{\text{RDCC}} = K_{ij} \left[1 - \chi_K \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (5)$$

The RDCC eigenvalue problem becomes

$$K^{\text{RDCC}} \mathbf{v}_k^{\text{RDCC}} = \lambda_k^{\text{RDCC}} \mathbf{v}_k^{\text{RDCC}}. \quad (6)$$

4 Spectral Signatures in RDCC

To first order in the relaxation parameter,

$$\lambda_k^{\text{RDCC}} = \lambda_k \left[1 - \chi_{\lambda,k} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (7)$$

RDCC therefore induces:

- enhanced large-scale eigenvalues (coherent modes),
- suppressed small-scale eigenvalues (relaxation damping),
- a characteristic turnover near ℓ_{rel} ,
- modified ordering of principal components,
- altered projection of persistence lifetimes onto spectral modes.

5 Principal Components in RDCC

The RDCC principal components are

$$\text{PC}_k^{\text{RDCC}}(D_i) = \sum_{j=1}^N v_{k,j}^{\text{RDCC}} K_{ij}^{\text{RDCC}}. \quad (8)$$

These components encode the dominant RDCC-modified topological modes.

6 Conclusion

Kernel PCA of persistence diagrams in the Relaxation-Driven Cyclic Cosmology reveals the spectral structure of linearized topology. The relaxation mechanism modifies the kernel eigenvalue spectrum in a scale-dependent manner, producing distinctive signatures in weak-lensing surveys. These effects provide a direct observational window into the relaxation scale and the temporal structure of the RDCC gravitational field. The present Companion establishes the theoretical foundation for future studies of spectral topology, kernel methods, and statistical tests in RDCC.

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